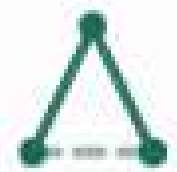


Track: **MICROFINANCE**

Problem Statement: **When One Borrower Falls Behind**



Kerdos

Microfinance Financial Contagion & Joint-Liability Risk Platform

Team Daemon

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Naman Goel

Riya M H

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PROBLEM STATEMENT

Problem Statement ID: P11

When One Borrower Falls Behind

A microfinance group can appear financially healthy even when one member begins experiencing financial stress. Shared guarantees, group lending structures, common income sources, and local economic conditions can connect borrowers, meaning that a problem affecting one member may create pressure on others. At the same time, some borrowers may simply be experiencing an isolated or temporary difficulty. The challenge is to understand when financial stress is contained and when it may become a wider group-level problem. Design a system that models how financial stress can develop and potentially propagate across a connected group of borrowers. The system should help distinguish isolated difficulty from vulnerability caused by another member and from independent deterioration in a borrower's own finances. Teams may explore how relationships are represented, how shocks are introduced and propagated, and how interventions could reduce wider risk without unnecessarily penalizing healthy borrowers. Historical or simulated borrower, repayment, income, and relationship data may be used.



Ingest & Input



Field Triage



Graph & Simulation



Portfolio Analytics

Edge Layer Visibility: All Relationship Edges (Full Network Mesh)

CASCADE REPLAY $t = 0$ of 11

PHASE 0: BASELINE STABILITY

Defaults (Red)

0 / 6

Stressed (Amber)

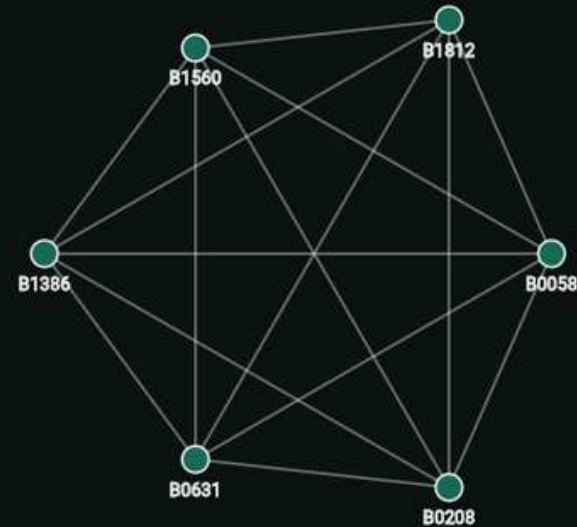
0

Peer Reserves

100%

Ledger: 100% On-Time (Blind Spot)

All members green. 100% peer reserves intact.



Healthy ($t=0$)

Stressed (Lagged)

Default Cascade (Peer Contagion)

12-Period Cascade Replay Scrubber

$t = 0$



Play Cascade



Custom Scenario & Shock Sandbox

Configure hypothetical stress scenarios to test peer contagion propagation and early-warning resilience.

Target Borrower:

B0058 (Primary Anchor • High Loan Exposure)

Stress Scenario Preset:

Medical Shock (Local)

Drought / Climate (Regional)

Price Inflation (Branch)

Custom Parameters

Injection Timestep:

$t = 2$

Shock Magnitude:

60% Deficit



Microfinance Financial Contagion & Joint-Liability Risk Platform

Select Workspace

Institutional Workspace
MFI Field Triage & Risk

Member Portal
Borrower & Group Circle

Work Email / Staff ID

officer@kerdos-mfi.org

Security PIN / Access Key

.....

Sign In to Institutional Workspace

Continue as Guest (Demo Mode)

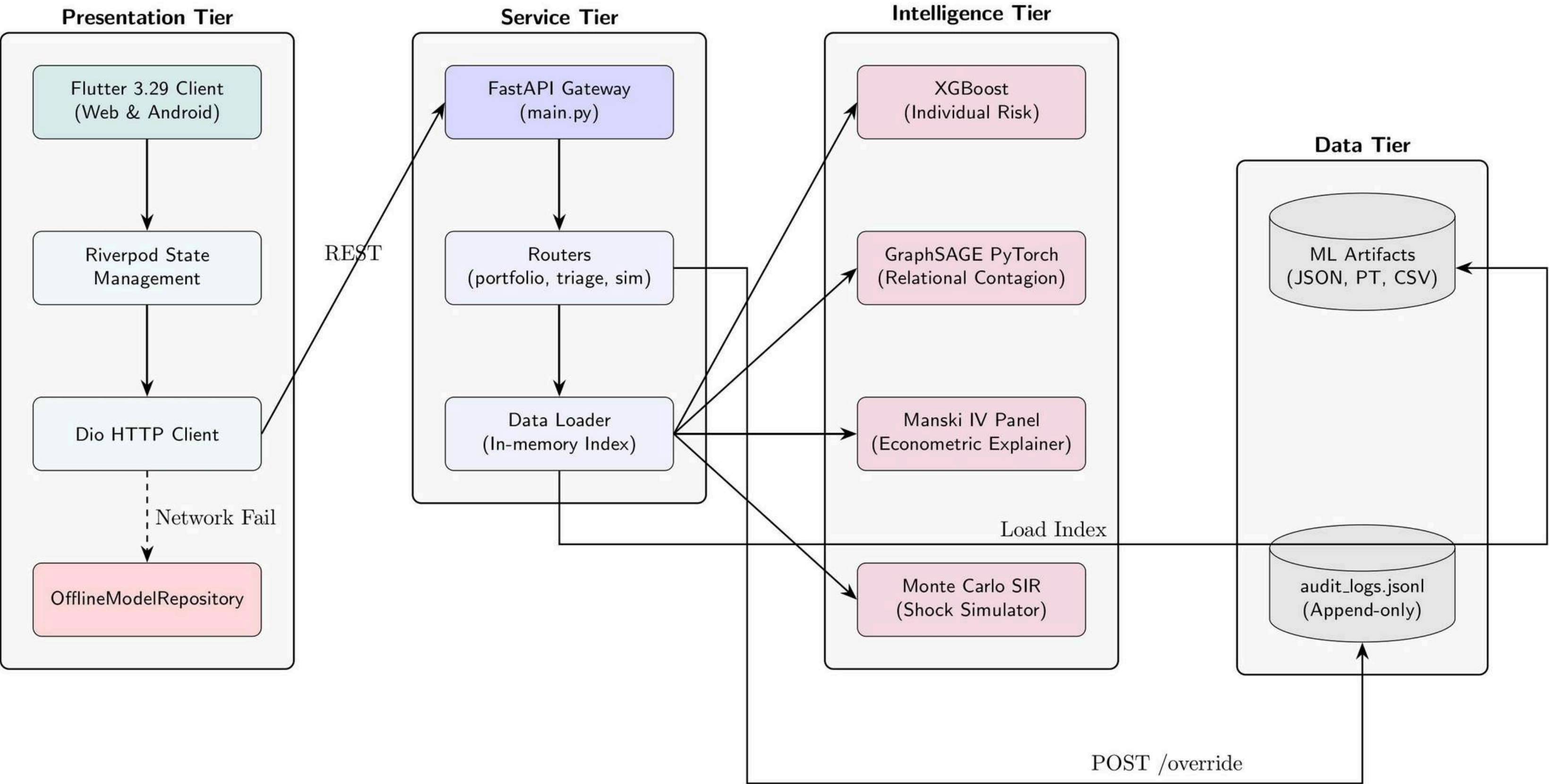
Made with by Team Daemon for M# 2026



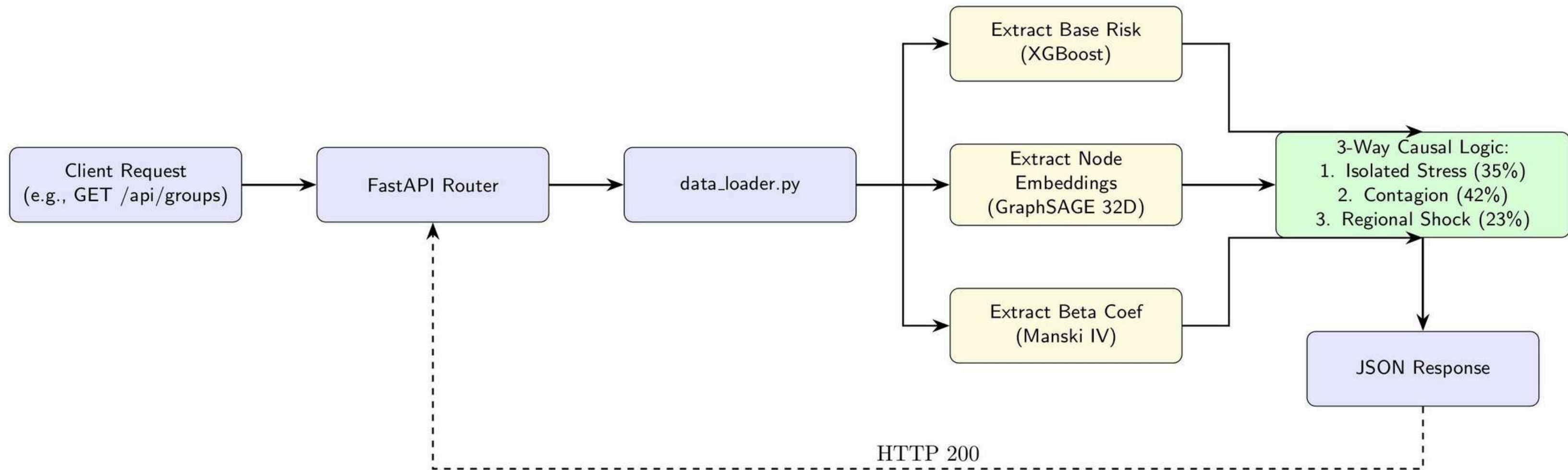
Microfinance Financial Contagion & Joint-Liability Risk Platform

Feature	Credit Bureaus (CIBIL/Equifax)	Legacy Core Banking	KERDOS (OUR APPLICATION)
Relationship Modeling	Isolated Individual	Static Group Flag	Dynamic Support Graph
Alert Speed	30–60 Days Late	Next-Day Batch	Real-Time Mobile Alert
Intervention Style	Punitive Score Cut	Full Group Freeze	Targeted Ringfence & Grace
Rural Connectivity	Web Portal Only	Clunky Online Mobile	100% Offline SQLite Sync

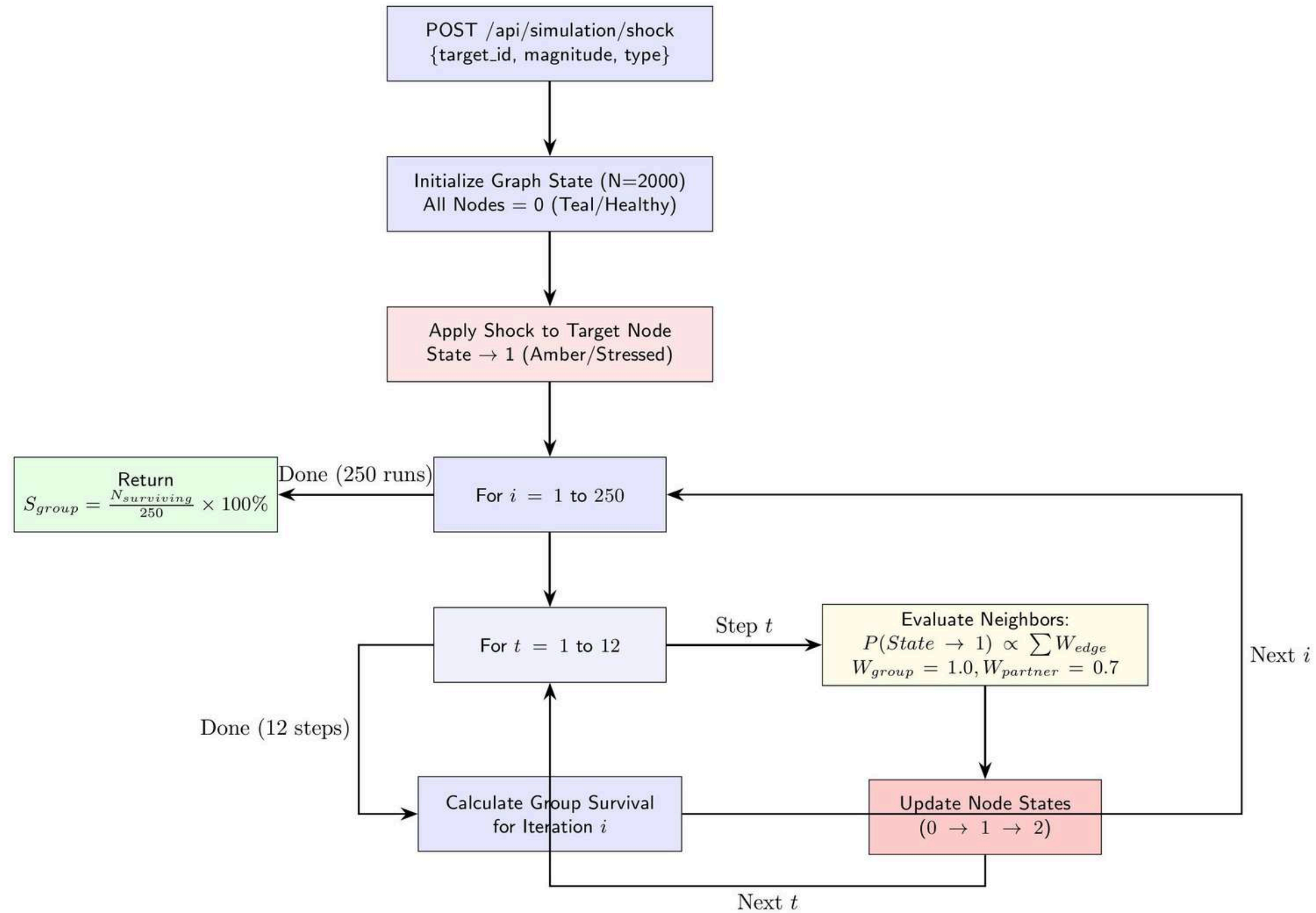
SYSTEM ARCHITECTURE



HIGH-LEVEL DESIGN (HLD)



LOW-LEVEL DESIGN (LLD)



MARKETING



1. Global Scale & Localization

Built for hyper-local deployment with native UI support across 8 regional languages (English, Hindi, Spanish, French, Swahili, Bengali, Marathi, Tamil), eliminating friction for rural field officers.



2. Target Market

Mid-to-large tier MFIs and rural banks across emerging economies (India, LATAM, Sub-Saharan Africa)



3. Operational Shift

Transitions MFI operations from reactive debt collection to proactive, data-driven loan restructuring.



 **Kerdos**

మైక్రోఫైనాన్స్ ఆర్థిక సంక్రమణ మరియు ఉమ్మడి బాధ్యత రిస్క్ ఫ్లాట్ ఫారమ్

వర్క్స్పేస్ను ఎంచుకోండి

 సంస్థాగత వర్క్స్పేస్
ఎంపిక పీల్డ్ క్రియాత్ & రిస్క్

 నిర్వహణ పోర్టల్
రుణగ్రహత & సమూహ సర్కిల్

కార్యాలయ ఇమెయిల్ / ఫ్లాట్ బాడి

 officer@kerdos-mfi.org

సెక్యూరిటీ పిన్ / యాక్సెస్ కీ



 సంస్థాగత వర్క్స్పేస్కు సైన్ ఇన్ చేయండి

 గెస్ట్గా కొనసాగండి (డెమో మోడ్)

రూపొందించబడింది ❤️ డి.ఎం. డెవలప్ బాపరా M# 2026 కోసం

B2B Enterprise Revenue Model

Tier 1: Core API Integration

- Headless integration directly into existing MFI MIS/CBS platforms.
- Priced per API call or per active borrower node modeled in the graph.

Tier 2: Enterprise Workspace SaaS

- Full deployment of the Kerdos Flutter application.
- Per-seat licensing for branch managers and field officers.
- Includes offline-first functionality and the Monte Carlo Intervention Sandbox.

Predictive Metrics & Simulation Efficacy

0.89	BASELINE AUROC
1.345	CONTAGION COEFFICIENT
34.0%	DOWNSTREAM RISK REDUCTION
< 4.0%	FALSE POSITIVE RATE
> 0.80	DEMOGRAPHIC PARITY RATIO

FURTHER IMPROVEMENTS & SCALING

- Integrate satellite anomalies into the regional layer to forecast agricultural stress 3 - 4 weeks before installments are missed.
- Upgrade static intervention simulations into self-learning policies that recommend the highest-yield relief action per borrower segment based on actual recovery data.
- Cross-MFI Federated Graph Learning will enable competing lenders to collaboratively detect hidden multi-borrower debt and cross-institution contagion.
- Better marketing techniques

A 3D paper cutout of a speech bubble is centered on a solid blue background. The speech bubble is white with a grey drop shadow, giving it a three-dimensional appearance. Inside the bubble, the words "THANK YOU!" are written in a bold, blue, sans-serif font. The entire scene is set against a light blue background with a subtle, abstract pattern of soft, overlapping shapes in various shades of blue and green.

THANK YOU!